

CHAPTER 1

PHYSICAL OPTICS— INTERFERENCE

INTRODUCTION

Wave theory of light attempts to understand the various optical phenomena exhibited by light waves. Interference constituted the first proof of the wave nature of light. Thomas Young first experimentally demonstrated interference in light waves. The superposition principle forms the conceptual basis for the explanation of interference. To produce interference, the light waves should be coherent, i.e., the light waves should have constant phase difference and same frequencies.

1.1 INTERFERENCE

When two or more light waves superimpose in the medium then according to superposition principle, the resultant displacement at any point is equal to the algebraic sum of the displacements due to individual waves. If y_1 and y_2 are the displacements due to two light waves 1 and 2, then the resultant displacement of the resultant wave is given by $y = y_1 + y_2$. The variation of resultant displacement influences amplitude variation, which causes intensity variations. This modification in the distribution of intensity in the region of superposition is known as *interference*. When the resultant amplitude is the sum of the amplitudes due to

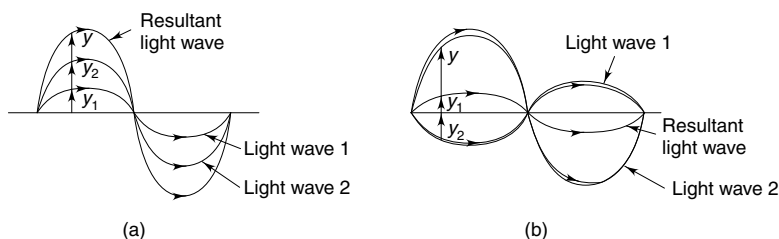


Figure 1.1 (a) Constructive interference ($y' = y_1 + y_2$);
(b) Destructive interference ($y = y_1 - y_2$)

two light waves, the interference is *constructive interference* (Fig. 1.1(a)). If the resultant amplitude is equal to the difference of two amplitudes, the interference becomes *destructive interference* (Fig. 1.1(b)). The intensity variations are studied as interference patterns or fringes or bright and dark fringes.

1.1.1 Coherence

To observe interference, the two light waves should be coherent. The two light waves are said to be coherent if they have the same frequency (wavelength) and they are in phase or maintain a constant phase relationship. When such light waves superimpose, they give rise to an interference pattern. The light waves emitted by two independent light sources do not have a constant phase relationship for a long time and they have randomly varying phase relationship. Due to which, they do not produce stationary interference pattern.

When a light from a single source is made to split into two wavefronts by using two slits, then these split wavefronts act as if they emerged from two sources having a fixed phase relationship and by superimposition, they produce a stationary interference pattern. The process of maintaining constant phase relation is known as coherence.

Coherence is of two types.

1. **Temporal Coherence** If it is possible to predict the phase relation at a point on the wave w.r.t. another point on the same wave then the wave has temporal coherence.
2. **Spatial Coherence** If it is possible to predict the phase relation at a point on a wave w.r.t. another point on a second wave, then the wave is said to have spatial coherence.

1.1.2 Conditions for Interference

1. The two light sources emitting light waves should be coherent.
2. The two sources must emit continuous light waves of same wavelengths and frequency.
3. The separation between the two sources should be small.
4. The distance between the two sources and the screen should be large.
5. To view interference fringes, the background should be dark.
6. The amplitudes of the light waves should be equal or nearly equal.
7. The sources should be narrow, i.e., they must be small.
8. The sources should be monochromatic.

1.1.3 Interference in Thin Film by Reflection

When light is incident on a plane parallel thin film, some portion gets reflected from the upper surface and the remaining portion is transmitted into the film. Again, some portion of the transmitted light is reflected back into the film by the lower surface and emerges through the upper surface. These reflected light

beams superimpose with each other, producing interference and forming interference patterns.

Consider a transparent plane parallel thin film of thickness ' t ' with refractive index μ . Let a monochromatic light ray AB be incident at an angle of incidence of ' i ' on the upper surface of the film. BE and BC are the reflected and transmitted light rays. Let the angle of refraction is ' r '. The ray BC will be reflected into the film and emerge through the film in the form of the light ray DF . These two light rays superimpose and depending upon path difference between them, they produce interference patterns.

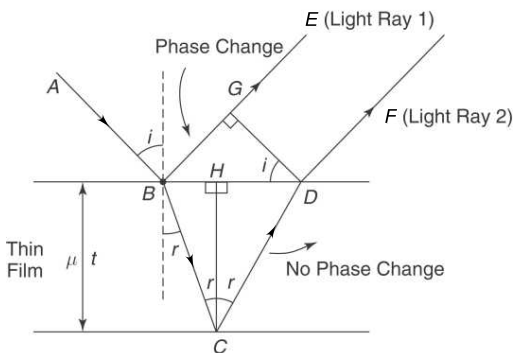


Figure 1.2 Interference in thin film

To know the path difference, let us draw the normal DG to BE . From the points D and G onwards, the light rays travel equal distances. By the time the light ray travels from B to G , the transmitted light ray has to travel from B to C and C to D .

The path difference between light rays (1) and (2) is

$$\text{Path difference} = \mu(BC + CD) \text{ in film} - BG \text{ in air} \quad (1)$$

Consider the $\triangle BCH$, $\cos r = \frac{HC}{BC}$

$$BC = \frac{HC}{\cos r} = \frac{t}{\cos r}$$

Similarly, from $\triangle DCH$, $CD = \frac{t}{\cos r}$

$$BC = CD = \frac{t}{\cos r} \quad (2)$$

To calculate BG , first BD which is equal to $(BH + HD)$ has to be obtained.

From $\triangle BHC$, $\tan r = \frac{BH}{CH} = \frac{BH}{t}$

$$BH = t \tan r$$

Similarly, $HD = t \tan r$

$$BD = BH + HD = 2t \tan r \quad (\because BH = HD) \quad (3)$$

From $\triangle BGD$, $\sin i = \frac{BG}{BD}$

$$BG = BD \sin i = 2t \tan r \sin i$$

From Snell's law, $\sin i = \mu \sin r$

$$BG = 2\mu t \tan r \sin r \quad (4)$$

Substituting the above values in Eq. (1)

$$\begin{aligned} \text{Path difference} &= \frac{2\mu t}{\cos r} - 2\mu t \tan r \sin r \\ &= \frac{2\mu t}{\cos r} - \frac{2\mu t \sin^2 r}{\cos r} \\ &= \frac{2\mu t}{\cos r} (1 - \sin^2 r) = \frac{2\mu t}{\cos r} \cos^2 r \\ &= 2\mu t \cos r \end{aligned} \quad (5)$$

At the point B , reflection occurs from the upper surface of the thin film (denser medium). Light ray (1) undergoes an additional phase change of π or an additional path difference of $\frac{\lambda}{2}$.

$$\text{Total path difference} = 2\mu t \cos r + \frac{\lambda}{2}$$

When the path difference is equal to integral multiples of λ then the rays (1) and (2) meet in phase and undergo constructive interference.

The condition for bright fringe is

$$\begin{aligned} 2\mu t \cos r + \frac{\lambda}{2} &= n\lambda \\ 2\mu t \cos r &= (2n - 1) \frac{\lambda}{2} \quad \text{where } n = 0, 1, 2, 3 \dots \end{aligned} \quad (6)$$

When the path difference is equal to half integral multiples of λ then the rays (1) and (2) meet in out of phase and undergo destructive interference.

The condition for dark fringe is

$$\begin{aligned} 2\mu t \cos r + \frac{\lambda}{2} &= (2n + 1) \frac{\lambda}{2} \\ 2\mu t \cos r &= n\lambda \quad \text{where } n = 0, 1, 2, 3 \dots \end{aligned} \quad (7)$$

Depending on the above conditions, the interference pattern consists of bright and dark fringes.

1.2 NEWTON'S RINGS

Newton's rings are one of the best examples for the interference in a nonuniform thin film. When a plano-convex lens with its convex surface is placed on a plane glass plate, an air film of increasing thickness is formed between the two. The thickness of the film at the point of contact is zero. If monochromatic light is allowed to fall normally and the film is viewed in the reflected light, alternate

dark and bright rings concentric around the point of contact between the lens and glass plate are seen. These circular rings were discovered by Newton and are called Newton's rings.

1.2.1 Experimental Arrangement

The experimental arrangement is as shown in Fig. 1.3. The plano-convex lens (L) of large radius of curvature is placed with its convex surface on a plane glass plate (P). The lens makes the contact with the plate at ' O '. The monochromatic light falls on a glass plate G held at an angle of 45° with the vertical. The glass plate G reflects normally a part of the incident light towards the air film enclosed by the lens L and the glass plate P . A part of the light is reflected by the curved surface of the lens L and a part is transmitted which is reflected back from the plane surface of the plate. These reflected rays interfere and give rise to an interference pattern in the form of circular rings. These rings are seen near the upper surface of the air film through the microscope.

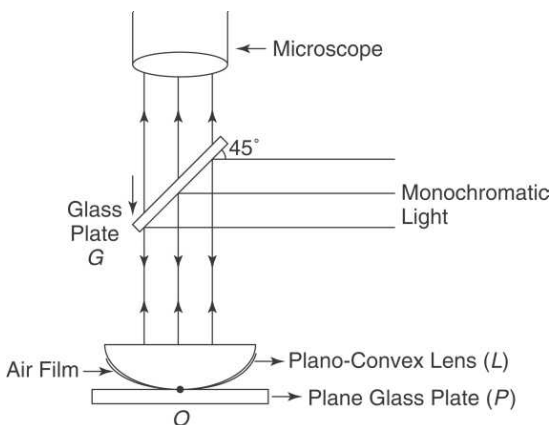


Figure 1.3 Newton's rings (Experimental set up)

and a part is transmitted which is reflected back from the plane surface of the plate. These reflected rays interfere and give rise to an interference pattern in the form of circular rings. These rings are seen near the upper surface of the air film through the microscope.

1.2.2 Explanation of Newton's Rings

Newton's rings are formed due to interference between the light rays reflected from the top and bottom surfaces of air film between the plate and the lens. The formation of Newton's rings can be explained with the help of Fig. 1.4. A part of the incident monochromatic light AB is reflected at B (glass–air boundary) in the form of the ray (1) with any additional phase (or path) change. The other part of the light is refracted along BC . Then at C (air–glass boundary), it is again reflected in the form of the ray (2) with additional phase

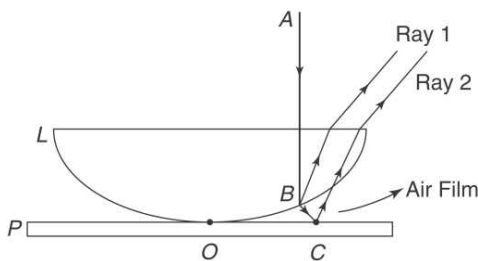


Figure 1.4 Formation of Newton's rings

change of π or path change of $\frac{\lambda}{2}$.

As the rings are observed in the reflected light, the path difference between them is $2\mu t \cos r + \frac{\lambda}{2}$. For air film $\mu = 1$ and for normal incidence $r = 0$, path difference is $2t + \frac{\lambda}{2}$.

At the point of contact $t = 0$, path difference is $\frac{\lambda}{2}$, i.e., the reflected light at the point of contact suffers phase change of π . Then the incident and reflected lights are out of phase and interfere destructively. Hence the central spot is dark.

The condition for bright ring is $2t + \frac{\lambda}{2} = n\lambda$.

$$2t = (2n - 1)\frac{\lambda}{2} \quad \text{where } n = 1, 2, 3 \dots$$

The condition for dark ring is

$$2t + \frac{\lambda}{2} = (2n + 1)\frac{\lambda}{2}$$

$$2t = n\lambda \quad \text{where } n = 0, 1, 2, 3 \dots$$

For monochromatic light, the bright and dark rings depend on the thickness of the air film. For a Newton's rings system, the locus of points having the same thickness lie on a circle having its centre at the point of contact. Thus, we get bright and dark circular rings with the point of contact as the centre.

1.2.3 Theory of Newton's Rings

To find the diameters of dark and bright rings, let 'L' be a lens placed on a glass plate P. The convex surface of the lens is the part of a spherical surface (Fig. 1.5) with centre at 'C'. Let R be the radius of curvature and r be the radius of Newton's ring corresponding to the film thickness 't'.

From the property of a circle,

$$NA \times NB = NO \times ND$$

Substituting the values,

$$r \times r = t \times (2R - t)$$

$$r^2 = 2Rt - t^2$$

As t is small, t^2 will be negligible

$$r^2 = 2Rt$$

$$t = \frac{r^2}{2R}$$

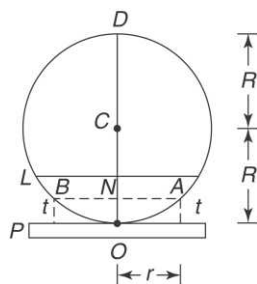


Figure 1.5 Theory of Newton's rings

For bright ring, the condition is

$$2t = (2n - 1) \frac{\lambda}{2}$$

$$2 \frac{r^2}{2R} = (2n - 1) \frac{\lambda}{2}$$

$$r^2 = \frac{(2n - 1)\lambda R}{2}$$

Replacing r by $\frac{D}{2}$, the diameter of n^{th} bright ring will be

$$\frac{D^2}{4} = \frac{(2n - 1)}{2} \lambda R$$

$$D = \sqrt{2n - 1} \sqrt{2\lambda R}$$

$$D \propto \sqrt{2n - 1}$$

$$D \propto \sqrt{\text{odd natural numbers}}$$

Thus the diameter of the bright rings are proportional to the square root of odd natural numbers. For dark ring, the condition is

$$2t = n\lambda$$

$$2 \frac{r^2}{2R} = n\lambda$$

$$r^2 = n\lambda R$$

$$D^2 = 4n\lambda R$$

$$D = 2 \sqrt{n\lambda R}$$

$$D \propto \sqrt{n}$$

$$D \propto \sqrt{\text{natural numbers}}$$

Thus, the diameters of dark rings are proportional to the square root of natural numbers.

With increase in the order (n), the rings get closer and the fringe width decreases and is shown in Fig. 1.6.

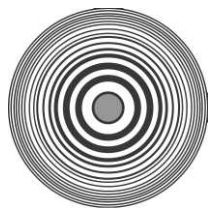


Figure 1.6 Newton's ring pattern

1.3 DETERMINATION OF WAVELENGTH OF A LIGHT SOURCE

The wavelength of a light source can be calculated by measuring the diameters of the various rings and knowing the radius of curvature of the plano convex lens.

Theory

Let R be the radius of curvature of a plano-convex lens, λ be the wavelength of light used. Let D_m and D_n are the diameters of m^{th} and n^{th} dark rings respectively. Then

$$D_m^2 = 4m\lambda R$$

and

$$D_n^2 = 4n\lambda R$$

$$D_n^2 - D_m^2 = 4(n - m)\lambda R$$

$$\lambda = \frac{D_n^2 - D_m^2}{4(n - m)R}$$

Using the above formula, λ can be calculated.

Experiment

Experimental arrangements are made as given in Section 1.2.1. After forming the Newton's rings, the microscope is adjusted so that the centre of the crosswire is adjusted at the central dark spot of the ring pattern. By counting the number of rings, the microscope is moved to the extreme left of the pattern and the cross wire is adjusted tangentially in the middle of the n^{th} (21st) dark ring. The reading of the microscope is noted.

Now the microscope is moved to the right and its readings are noted successively at $(n - 3)^{\text{th}}$ (18th), $(n - 6)^{\text{th}}$ (15th) ... rings,

etc., with a difference of three rings up to the central dark spot. Again crossing the central dark spot in the same direction, the readings corresponding to $(n - 6)^{\text{th}}$ (15th), $(n - 3)^{\text{th}}$ (18th) and n^{th} (21st) rings are noted. The difference between the left and right reading gives the diameter of the particular ring. A graph is plotted between the number of the rings and the square of the corresponding diameter.

From the graph,

$$\frac{D_n^2 - D_m^2}{n - m} = \frac{AB}{CD}$$

The radius R of the plano convex lens can be obtained with the help of a spherometer. Substituting these values in the formula, the wavelength of the light source can be known.

1.4 DETERMINATION OF REFRACTIVE INDEX OF A LIQUID

The experiment is performed when there is an air film between the glass plate and the plano-convex lens. The diameters of m^{th} and n^{th} dark rings are determined with the help of a travelling microscope. We have

$$D_n^2 - D_m^2 = 4(n - m)\lambda R \quad (8)$$

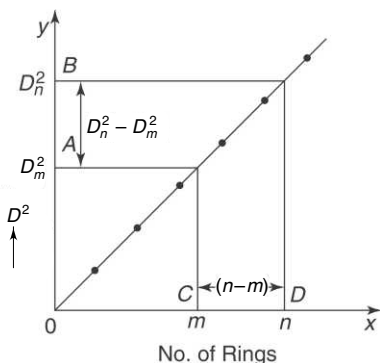


Figure 1.7 Plot of D^2 w.r.t. the number of rings

The plano convex lens and plane glass plate is placed into the container which consists of the liquid whose refractive index (μ) is to be determined. Now, the air film is replaced by the liquid film. Again, the diameters of the same m^{th} and n^{th} dark rings are to be obtained. Then we have

$$D_n'^2 - D_m'^2 = \frac{4(n-m)\lambda R}{\mu} \quad (9)$$

From equations (8) and (9), we get

$$\mu = \frac{D_n^2 - D_m^2}{D_n'^2 - D_m'^2} \quad (10)$$

Using the above formula, ' μ ' can be calculated.

SOLVED PROBLEMS

1. A parallel beam of light of $6000\text{\AA}$ is incident on a thin glass plate of refractive index 1.5 such that the angle of refraction into the plate is 50° . Find the least thickness of the glass plate which will appear dark by reflection.

Given data Wavelength of the given light $\lambda = 6000\text{\AA} = 6000 \times 10^{-8} \text{ cm}$
 Refractive index of the glass plate $\mu = 1.5$
 Angle of refraction $r = 50^\circ$

Solution $2\mu t \cos r = n\lambda$

For least thickness, $n = 1$

$$\text{Thickness } t = \frac{n\lambda}{2\mu \cos r}$$

$$t = \frac{1 \times 6000 \times 10^{-8}}{2 \times 1.5 \times \cos 50^\circ}$$

$$t = \frac{6 \times 10^{-5}}{3 \times 0.6428} = 3.1114 \times 10^{-5} \text{ cm}$$

$$= 3.11 \times 10^{-5} \text{ cm}$$

2. In a Newton's rings experiment, the diameter of the 5^{th} ring is 0.30 cm and the diameter of the 15^{th} ring is 0.62 cm. Find the diameter of the 25^{th} ring.

Given data Diameter of the 5^{th} ring = 0.3 cm
 Diameter of the 15^{th} ring = 0.62 cm

Solution We want diameter of the 25th ring.

We know that $D_n^2 - D_m^2 = 4(n - m)\lambda R$

$$D_{15}^2 - D_5^2 = 4(15 - 5)\lambda R$$

$$\Rightarrow (0.62)^2 - (0.3)^2 = 40 \lambda R$$

$$0.3844 - 0.09 = 40 \lambda R \quad (i)$$

Similarly, D_{25}^{th} and D_5^{th}

$$D_{25}^2 - D_5^2 = 4(25 - 5)\lambda R$$

$$D_{25}^2 - 0.09 = 80\lambda R \quad (ii)$$

Dividing Eq. (ii) by Eq. (i)

$$\frac{D_{25}^2 - 0.09}{0.3844 - 0.09} = \frac{80 \lambda R}{40 \lambda R}$$

$$\Rightarrow D_{25}^2 = 2 \times 0.2944 + 0.09 = 0.6788$$

$$D_{25} = \sqrt{0.6788} = 0.824 \text{ cm}$$

Diameter of the 25th ring is 0.824 cm.

3. A convex lens on a plane glass plate is exposed to a monochromatic light. The diameter of the 10th dark ring is 0.433 cm. Find the wavelength of the light used if the radius of curvature of the lens is 70 cm.

Given data Radius of curvature, $R = 70 \text{ cm}$

Diameter of the 10th ring = 0.433 cm

Solution

$$\lambda = \frac{D_n^2}{4nR}$$

$$= \frac{(0.433)^2}{4 \times 10 \times 70} = \frac{0.187}{2800} = 6.6786 \times 10^{-5} \text{ cm}$$

4. A soap film with a refractive index of 1.33 and thickness of 5000 Å is exposed to white light. What wavelengths in the visible region are reflected?

Given data Refractive index of the soap film, $\mu = 1.33$

Thickness of the film, $t = 5000 \text{ Å} = 5000 \times 10^{-10} \text{ m}$

Solution Condition for maximum reflection is

$$2\mu t \cos r = (2n + 1) \frac{\lambda}{2}, \quad \text{where } n = 0, 1, 2, \text{ etc.}$$

Assume $\cos r = 1$

Then

$$\lambda = \frac{4\mu t}{(2n + 1)} = \frac{4 \times 1.33 \times 5000 \times 10^{-10}}{(2n + 1)}$$

$$= \frac{26600 \times 10^{-10}}{(2n + 1)} \text{ m}$$

When $n = 0$, $\lambda_1 = 26600 \times 10^{-10} \text{ m}$ (infrared region)

When $n = 1$, $\lambda_2 = \frac{26600 \times 10^{-10}}{3} = 8866.7 \times 10^{-10} \text{ m}$ (I R)

$n = 2$, $\lambda_3 = \frac{26600 \times 10^{-10}}{5} = 5320 \times 10^{-10} \text{ m}$ (visible region)

$n = 3$, $\lambda_4 = \frac{26600 \times 10^{-10}}{7} = 3800 \times 10^{-10} \text{ m}$

(ultraviolet region)

All wavelengths reflected, at wavelength $5320 \text{ \AA}$ the visible regions are reflected.

5. What is the thickness of the thinnest film of 1.33 refractive index in which destructive interference of the yellow light ($6000 \text{ \AA}$) of a normally incident beam in air can take place by reflection?

Given data Refractive index of the film, $\mu = 1.33$

Wavelength of yellow light $\lambda = 6000 \text{ \AA} = 6000 \times 10^{-8} \text{ cm}$

Angle of incidence $i = 0^\circ$, $\therefore r = 0^\circ$ (reflection)

For least thickness, $n = 1$

Solution

$$2\mu t \cos r = n\lambda$$

$$t = \frac{n\lambda}{2\mu \cos r}$$

$$= \frac{1 \times 6000 \times 10^{-8}}{2 \times 1.33 \times \cos 0^\circ}$$

$$= \frac{6 \times 10^{-5}}{2.66 \times 1} = 2.255 \times 10^{-5} \text{ cm}$$

$\therefore$ the thickness of the thinnest film $= 2.255 \times 10^{-5} \text{ cm}$

6. In a Newton's rings experiment, the diameter of the 8th ring was 0.35 cm and the diameter of the 18th ring was 0.65 cm. If the wavelength of the light used is $6000 \text{ \AA}$ then find the radius of curvature of the plano-convex lens.

Given data

Diameter of the 18th ring $= 0.65 \text{ cm}$

Diameter of the 8th ring $= 0.35 \text{ cm}$

Wavelength of the light, $\lambda = 6000 \text{ \AA}$

Solution

Radius of curvature of the plano-convex lens is

$$R = \frac{D_n^2 - D_m^2}{4(n - m)\lambda}$$

$$\Rightarrow R = \frac{(0.65)^2 - (0.35)^2}{4(18 - 5) \times 600 \times 10^{-8}}$$

$$= \frac{0.4225 - 0.1225}{24 \times 10^{-4}} = \frac{0.3}{24} \times 10^4 = 125 \text{ cm}$$

$\therefore$ radius of curvature of the plano-convex lens is 125 cm.

7. In a Newton's rings experiment, the diameter of the 12th ring changes from 1.45 cm to 1.25 cm when a liquid is introduced between the lens and the glass plate. Find the refractive index of the liquid.

Given data Diameter of the 12th ring in air medium, $D_{12} = 1.45 \text{ cm}$
 Diameter of the 12th ring in a liquid, $D'_{12} = 1.25 \text{ cm}$

Solution

$$D_n^2 = \frac{4n\lambda R}{\mu}$$

$$(D_{12})^2 = \frac{4n\lambda R}{\mu} = \frac{4 \times 12\lambda R}{1} \quad \therefore \mu = 1 \text{ for air}$$

$$(D'_{12})^2 = \frac{4 \times 12\lambda R}{\mu}$$

From the above relation,

$$\mu = \frac{(D_{12})^2}{(D'_{12})^2} = \frac{(1.45)^2}{(1.25)^2} = \frac{2.1025}{1.5625} = 1.3456$$

$\therefore$ refractive index of the liquid is 1.3456.

8. In a Newton's ring experiment, the diameter of the 5th dark ring is 0.3 cm and the diameter of the 25th ring is 0.8 cm. If the radius of curvature of the plano convex lens is 100 cm, find the wavelengths of the light used.

Given Data Diameter of 5th dark ring $D_5 = 0.3 \text{ cm}$
 Diameter of 25th dark ring $D_{25} = 0.8 \text{ cm}$
 Radius of curvature of plano convex lens $R = 100 \text{ cm}$

Solution

$$\lambda = \frac{D_n^2 - D_m^2}{4(n - m)R}$$

$$= \frac{(0.8)^2 - (0.3)^2}{4(8 - 3) \times 100}$$

$$= 4.87 \times 10^{-5} \text{ cm}$$

MULTIPLE CHOICE QUESTIONS

1. Two sources are said to be coherent if their emitted waves have
 - (a) same amplitude
 - (b) same wavelength
 - (c) constant phase difference
 - (d) All of these
2. When a light wave is reflected at the glass–air interface, the change of phase of the reflected wave is equal to
 - (a) 0
 - (b) $\pi/2$
 - (c) $\pi/4$
 - (d) π
3. When a light wave is reflected at the air–glass interface, the change of phase of the reflected wave is equal to
 - (a) 0
 - (b) $\pi/2$
 - (c) $\pi/4$
 - (d) π
4. In the presence of a plane parallel film, the path difference is equal to
 - (a) $2\mu \sin i$
 - (b) $2\mu t \cos r$
 - (c) $2\mu \sin r$
 - (d) $2\mu t \cos i$
5. If a light wave is refracted from air to denser medium then the phase and path difference is
 - (a) π and λ
 - (b) π and $\lambda/2$
 - (c) $\pi/2$ and λ
 - (d) $\pi/2$ and $\lambda/2$
6. In a plane parallel film due to reflected light, what is the condition for maxima for the air film to appear bright?
 - (a) $2\mu t \cos r = n\lambda$
 - (b) $2\mu t \cos r = (2n + 1)\lambda/2$
 - (c) $2\mu t \cos r = n\lambda/2$
 - (d) $2\mu t \cos r = (2n - 1)\lambda/2$
7. In a plane parallel film due to reflected light, what is the condition for minima for the air film to appear dark?
 - (a) $2\mu t \cos r = n\lambda/2$
 - (b) $2\mu t \cos r = (2n + 1)\lambda/2$
 - (c) $2\mu t \cos r = n\lambda$
 - (d) $2\mu t \cos r = (2n - 1)\lambda/2$
8. In Newton's rings, interference is due to light rays reflected from
 - (a) upper surface of lens and upper surface of glass plate
 - (b) lower surface of lens and upper surface of glass plate
 - (c) lower surface of glass plate and upper surface of lens
 - (d) lower surface of lens and lower surface of glass plate
9. In Newton's rings experiment, what is the condition for bright fringes in case of reflected light?
 - (a) $D \propto \sqrt{2n + 1}$
 - (b) $D \propto n$
 - (c) $D \propto \sqrt{n}$
 - (d) $D \propto \sqrt{2n - 1}$

10. In Newton's rings experiment, what is the condition for dark fringes in case of reflected light?
- (a) $D \propto \sqrt{2n+1}$ (b) $D \propto n$
(c) $D \propto \sqrt{n}$ (d) $D \propto \sqrt{2n-1}$
11. If the thickness of the parallel thin film increases, the path difference between the reflected lights
- (a) decreases
(b) increases
(c) first decreases and then increases
(d) first increases and then decreases
12. The phenomenon of intensity modification due to superposition of light waves is known as
- (a) interference (b) diffraction
(c) polarisation (d) coherence
13. When a light wave is reflected at the air-glass interface, the change of path of the reflected wave is equal to
- (a) $\lambda/2$ (b) $\lambda/4$
(c) $\lambda/6$ (d) $\lambda/8$
14. When a liquid film is introduced in the place of air film, then the diameters of the rings
- (a) increases
(b) decreases
(c) first increases and then decreases
(d) first decreases and then increases
15. When the film is observed by white light, colours are observed due to
- (a) interference (b) diffraction
(c) polarisation (d) coherence
16. Wave nature of light is evidenced by
- (a) photoelectric effect (b) interference
(c) blackbody radiation (d) nuclear emission
17. In the interference pattern energy is
- (a) created at the maxima (b) destroyed at the minima
(c) converted but is redistributed (d) converted into matter
18. Two waves having their intensities in the ratio 9 : 1 produce interference. In the interference pattern, the ratio of maximum to minimum intensity is equal to
- (a) 2 : 1 (b) 9 : 1 (c) 3 : 1 (d) 4 : 1
19. Two beams interfere have their amplitudes ratio 2 : 1. Then the intensity ratio of bright and dark fringes is
- (a) 2 : 1 (b) 1 : 2 (c) 9 : 1 (d) 4 : 1
20. When a thin film of oil or soap bubble is illuminated with white light, multiple colours appear. This is due to

- (a) diffraction (b) polarisation
(c) total internal reflection (d) interference
21. When the Newton's rings are viewed through a microscope, the Central ring is seen dark because
(a) Phase difference of π due to phase change on reflection.
(b) Air film thickness is nearer to zero.
(c) Path difference of λ due to phase change on reflection.
(d) Phase change of 2π due to phase change on reflection.
22. The convex lense in Newton's ring apparatus is replaced by an ordinary glass plate, then
(a) non-interference occurs
(b) circular rings are still obtained
(c) interference takes place but the shape of fringes is irregular
(d) straight line fringes are observed.
23. In Newton's ring experiment the diameter of 40th ring is 0.1 m air film. When an oil film is formed the diameter of the same ring becomes 0.089 m the refractive index of oil is
(a) 1.26 (b) 0.80 (c) 1.10 (d) 1.33

Answers

- | | | | | | |
|---------|---------|---------|---------|---------|---------|
| 1. (d) | 2. (a) | 3. (d) | 4. (b) | 5. (b) | 6. (d) |
| 7. (c) | 8. (b) | 9. (a) | 10. (c) | 11. (b) | 12. (a) |
| 13. (a) | 14. (b) | 15. (a) | 16. (b) | 17. (c) | 18. (d) |
| 19. (c) | 20. (d) | 21. (a) | 22. (c) | 23. (a) | |

Fill in the Blanks

- The resultant displacement is due to the presence of both the waves of displacements y_1 and y_2 is given by $y = y_1 + y_2$. This is known as the principle of _____.
- Two waves are said to be _____ if their waves have same ratio of bright fringe to dark fringe.
- If 5 : 1 is the amplitude ratio of the source, the intensity ratio of maximum to minimum is
(a) 25 : 1 (b) 6 : 4
(c) 4 : 9 (d) 9 : 4
- Due to interference in thin films, if the film thickness is extremely small when compared to λ , the film will appear _____.
- Newton's ring experiment, the radius of n^{th} dark ring is given by _____ where λ is the wavelength of light and R is the radius of the ring observed.
- On introducing the liquid in between the lense and the glass plate in Newton's ring experiment, the diameter of the rings observed _____.

7. Superposition of two coherent wavefronts originating from coherent sources result in _____.
8. In interference pattern fringes have _____ width.

Answers to Fill in the blanks

- | | | |
|------------------|-----------------------|--------------|
| 1. Superposition | 2. Coherent | 3. 9:4 |
| 4. dark | 5. $\sqrt{nR\lambda}$ | 6. decreases |
| 7. Interference | 8. equal | |

True or False

1. Two monochromatic sources of same amplitude and same wavelength can act as coherent sources. (T/F)
2. If the path difference between the two interfering waves is integral multiple of λ , constructive interference occurs. (T/F)
3. In Newton's ring experiment reflected light the point of contact of lens with the glass plate. Where thickness of air is zero, appears dark. (T/F)
4. Radius of fourth dark ring in Newton's ring experiment is $4\sqrt{R\lambda}$. (T/F)
5. Thin films when viewed under white light, appears multicolored due to dispersion. (T/F)

Answers to True or False

- | | | | | |
|------|------|------|------|------|
| 1. T | 2. T | 3. T | 4. F | 5. F |
|------|------|------|------|------|

REVIEW QUESTIONS

Short Answer Questions

1. What is interference?
2. What is constructive interference?
3. What is destructive interference?
4. What is coherence?
5. What is temporal coherence?
6. What is spatial temporal coherence?
7. Mention any two important conditions to get interference.
8. Mention the conditions to get bright and dark fringes in the interference due to reflected light by thin film.
9. Why the centre spot is dark in the Newton's rings formed by the reflected light?
10. Why does Newton's rings consist of concentric rings?
11. Mention the expression for diameter of bright and dark rings in Newton's rings by the reflected light.

Essay Type Questions

1. Explain the interference in the films by reflection.
2. Describe the formation of Newton's rings with necessary theory.
3. Explain how the wavelength of light sources is determined by forming Newton's rings.
4. Explain how the refractive index of liquid is determined by forming Newton's rings.